

*** ES-X: Universal Guide for Theme Creators**

This document explains the universal extensions introduced in ES-X that allow any theme creator to design dynamic, configurable, multi-layout themes without depending on a specific visual style (PS3, PS4, Mini, ArtBook, etc.).

The purpose of this guide is:

- To provide public documentation for RetroPie theme developers.
- To help users experiment with layouts and dynamic themes.
- To establish a clean and universal foundation for theme.ini support.

This guide does NOT describe any style, graphics or assets from Pi-Station-X. It only documents the universal system available in ES-X.

1. System Overview

ES-X expands EmulationStation with these core mechanisms:

- Universal XML variables: `${variable}`
- A theme.ini: file defining layouts and values
- Theme Options allowing users to switch layouts without editing XML

How it works:

1. The theme XML contains placeholders like `${bgMain}`
2. The user selects a layout from Theme Options
3. ES-X loads the corresponding section from theme.ini
4. Every placeholder is replaced dynamically

This system makes themes modular, maintainable and easier to extend.

2. Universal Variables Available

Background and Overlay

`bgMain` → main background image

`bgOverlay` → decorative overlay

Help System

`systemHelp` → help box image

`systemHelpPos` → position X Y (0-1 normalized)

`systemHelpSize` → size W H (0-1 normalized)

Sounds

`scrollSoundList` → scroll sound in BASIC view

`scrollSoundDetailed` → scroll sound in DETAILED/VIDEO view

Fonts

`playersFont` → custom font for `md_players` field

System Logo

logoPosSystem → X Y position

2.1. Carousel Logo Parameters (Experimental / Upcoming)

Some theme creators may be familiar with Batocera's carousel system, which includes visual parameters such as opacity, spacing and scaling.

ES-X introduces a simplified, universal version of these parameters so they can be used in any theme.

These variables are optional and safe to use.

Themes that don't use them will simply fall back to default values.

Logo Opacity

logoOpacity → opacity of non-selected logos (0.0–1.0)

logoOpacityFocus → opacity of the focused logo

minLogoOpacity → minimum fade applied to far-side logos

Logo Scaling

logoScale → base scale for all carousel logos

logoScaleFocus → scale applied to the centered logo

logoScaleMin → minimum scale applied to distant logos

Logo Spacing

logoSpacing → base spacing between carousel logos

scaledLogoSpacing → additional spacing when scaling is active

Carousel Alignment

logoAlignment → left | center | right

These parameters allow:

- clearer visual hierarchy
- better readability with large logo sets
- compatibility with wide or tall layouts

This makes the carousel more flexible for PS-style themes, ArtBook-style minimal layouts and multi-system setups.

2.2. Text Border / Text Outline (Universal Feature)

ES-X supports the same text outline properties available in modern EmulationStation themes.

These parameters allow creators to add a clean border around text, improving readability against bright or complex backgrounds.

This feature is optional and can be used in any XML element of type <text>.

Available properties

textColor → main color of the text (hex)

textBorderColor → color of the outline/border (hex)

textBorderSize → thickness of the outline (0.0 – 1.0 normalized)

Example inside a <text> element

```
<text name="systemInfo">
<pos>0.12 0.05</pos>
<color>FFFFFF</color>
<textBorderColor>000000</textBorderColor>
<textBorderSize>0.015</textBorderSize>
<fontSize>0.03</fontSize>
<text>My System Info</text>
</text>
```

Notes

textBorderSize uses the same 0–1 normalized scale as other size values.

Borders are rendered outside the text glyph to avoid distortion.

Light text with dark outlines is recommended for maximum legibility (e.g. white + black border).

Works in SystemView, Gamelist Views, HelpSystem, and all <text> UI elements.

This feature helps replicate the clean readability found in ArtBook, ES-DE, and PS-style themes.

3. Counters Exposed by ES-X

The following system-wide counters can be displayed in any theme:

```
${gameCount}
${favoriteCount}
${mostPlayedCount}
${mostPlayedName}
```

```
${recentCount}
```

Metadata of the selected game:

```
${md_title}
${md_description}
${md_year}
${md_publisher}
${md_developer}
${md_players}
${md_rating}
${md_playcount}
${md_lastplayed}
${md_favorite}
```

All values are safe to use and universally available.

4. Universal Options for theme.ini

Enable/Disable Elements

showGameCount = true/false
showFavoriteCount = true/false
showMostPlayedCount = true/false
showRecentCount = true/false

Normalized Positions

gameCountPos = x y
favoriteCountPos = x y
mostPlayedCountPos = x y
recentCountPos = x y

Text Size and Color

gameCountSize = 0.02
gameCountColor = FFFFFFFF

5. Example theme.ini

```
[themeoptions]
name = "Layout Settings"
autoapply = true

[layout_smd]
bgMain = ./layout/SMD/main1.png
bgOverlay = ./layout/SMD/main2.png
systemHelp = ./layout/SMD/help.png
systemHelpPos = 0.68 0.78
systemHelpSize = 0.27 0.17

scrollSoundList = ./layout/SMD/sound_list.wav
scrollSoundDetailed = ./layout/SMD/sound_detailed.wav
```

Universal counters

showGameCount = true
showFavoriteCount = true
showMostPlayedCount = true

Positions

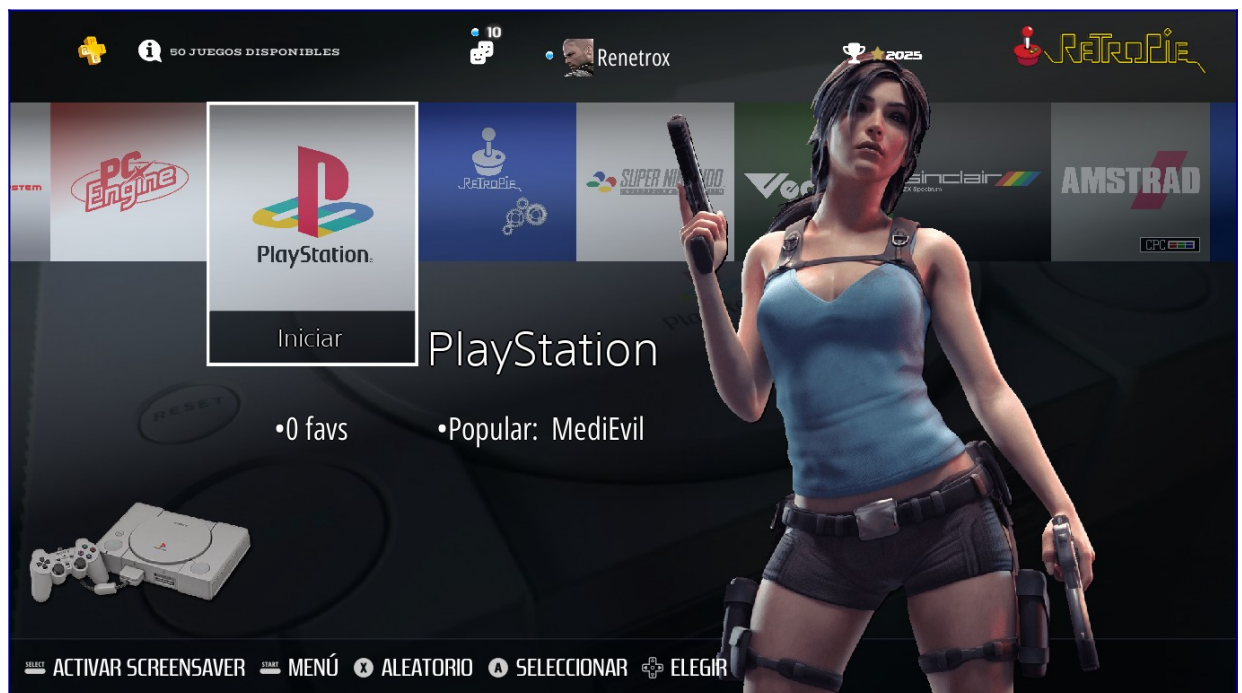
gameCountPos = 0.10 0.95
favoriteCountPos = 0.25 0.95
mostPlayedCountPos = 0.40 0.95

6. Example XML Implementation

```
<text name="gamecount" extra="true">
<pos>${gameCountPos}</pos>
<color>${gameCountColor}</color>
<fontSize>${gameCountSize}</fontSize>
<text>${gameCount}</text>
</text>

<text name="favoritecount" extra="true">
<pos>${favoriteCountPos}</pos>
<color>${favoriteCountColor}</color>
<fontSize>${favoriteCountSize}</fontSize>
<text>${favoriteCount}</text>
</text>

<text name="mostplayedcount" extra="true">
<pos>${mostPlayedCountPos}</pos>
<color>${mostPlayedCountColor}</color>
<fontSize>${mostPlayedCountSize}</fontSize>
<text>${mostPlayedCount}</text>
</text>
```



This XML block works on any theme.

7. About ArtBook

ES-X follows a modular design philosophy partially inspired by the clean and well-structured layout approach of the [b]ArtBook[/b] theme.

This mention is important because many theme creators already know ArtBook as a reference for:

- normalized positions
- separation of visual layers
- simple, flexible structure

ES-X does NOT copy or use any ArtBook graphics or assets.

Only the design philosophy and organizational approach influenced this universal system.

example:

```
; =====
; Art Book Next - Theme Options
; ES-X Compatible Format
; =====
```

```
;-----
; Aspect Ratio (aspectRatio)
```

```
;-----
colorScheme = steam-deck
gameListStyle = metadata-on-miximage
soundSet = es-de
aspectRatio = 4-3
```

```
[aspectRatio]
type = select
label = Aspect Ratio
apply_to = aspectRatio
values = 16-9 | 16:9, 16-10 | 16:10, 4-3 | 4:3
default = 16-9
```

```
;-----
; Game List Style (gameListStyle)
```

```
;-----
[gameListStyle]
type = select
label = Game List Style
apply_to = gameListStyle
values = metadata-on | Metadata ON, metadata-off | Metadata OFF, metadata-off-zoomed |
Metadata OFF (Zoom), metadata-on-miximage | Metadata + Miximage
default = metadata-on
```

```
;-----
; Color Scheme (colorScheme)
```

```
;-----
[colorScheme]
type = select
label = Color Scheme
apply_to = colorScheme
values = art-book-next | Art Book Next, art-book | Art Book, snes | SNES, famicom | Famicom,
```

steam-deck|Steam Deck, custom|Custom

default = art-book-next

;-----

; Sound Set (soundSet)

;-----

[soundSet]

type = select

label = Theme Sounds

apply_to = soundSet

values = es-de|ES-DE, steam-os|Steam OS

default = steam-os

;-----

; Extra profiles (optional)

; Works like your [layout_smd], [layout_psx].

; You can later create:

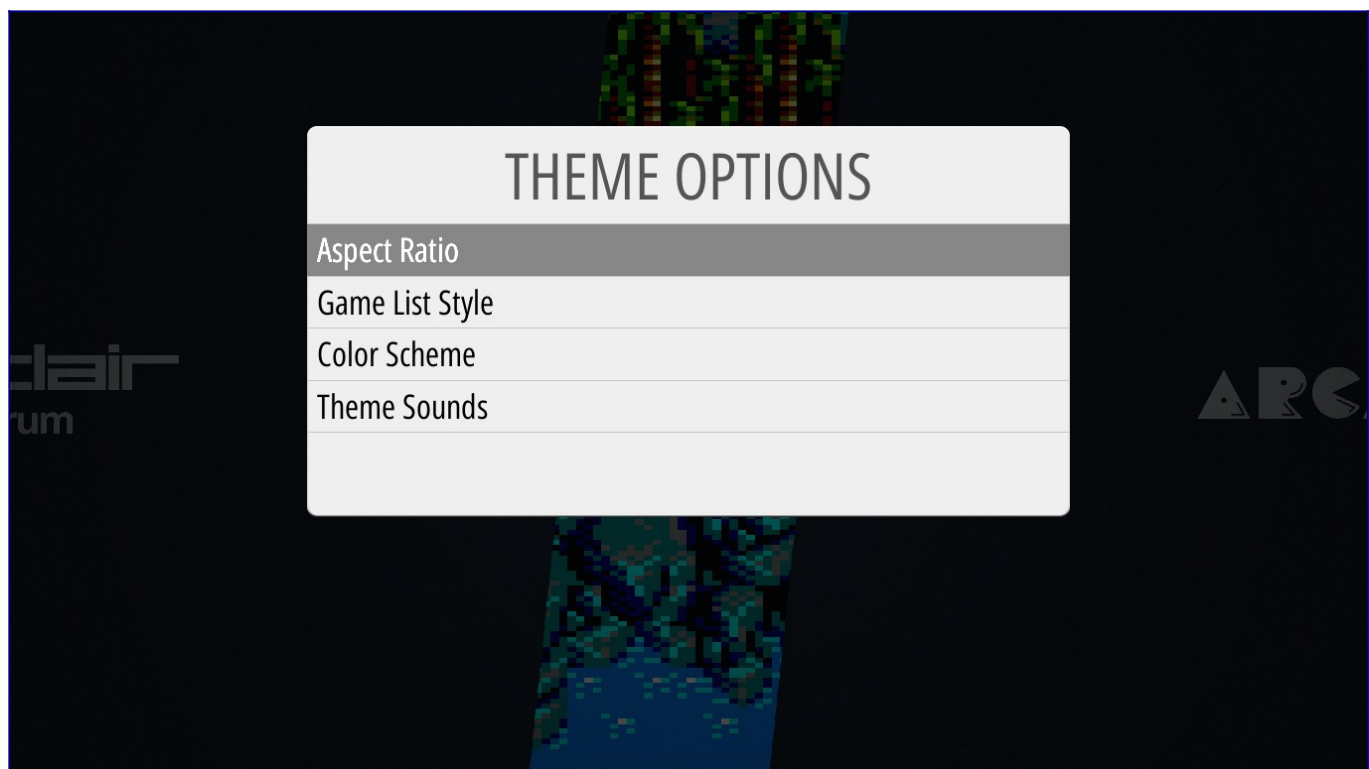
;

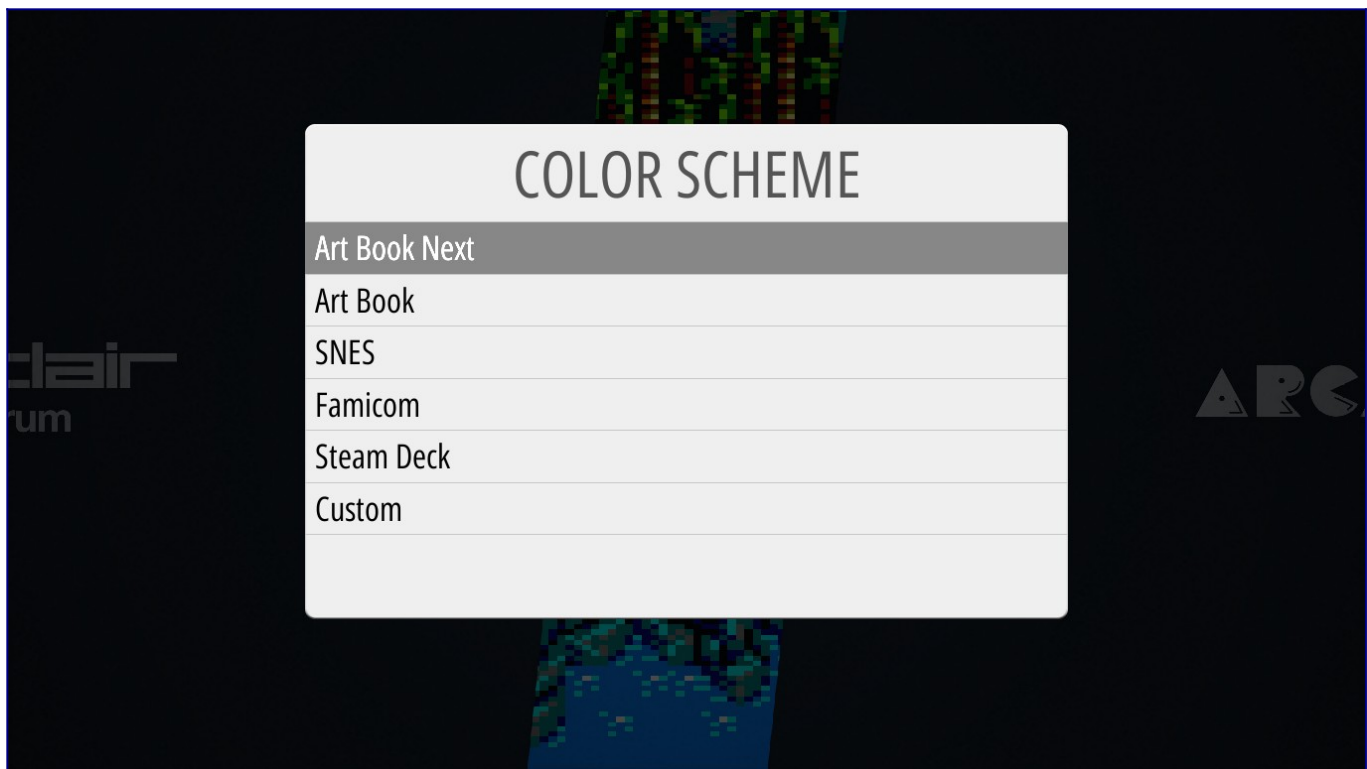
; [colorScheme_snes]

; [colorScheme_famicom]

; etc.

;-----





This guide is intended for anyone interested in experimenting with dynamic, multi-layout themes on RetroPie and EmulationStation